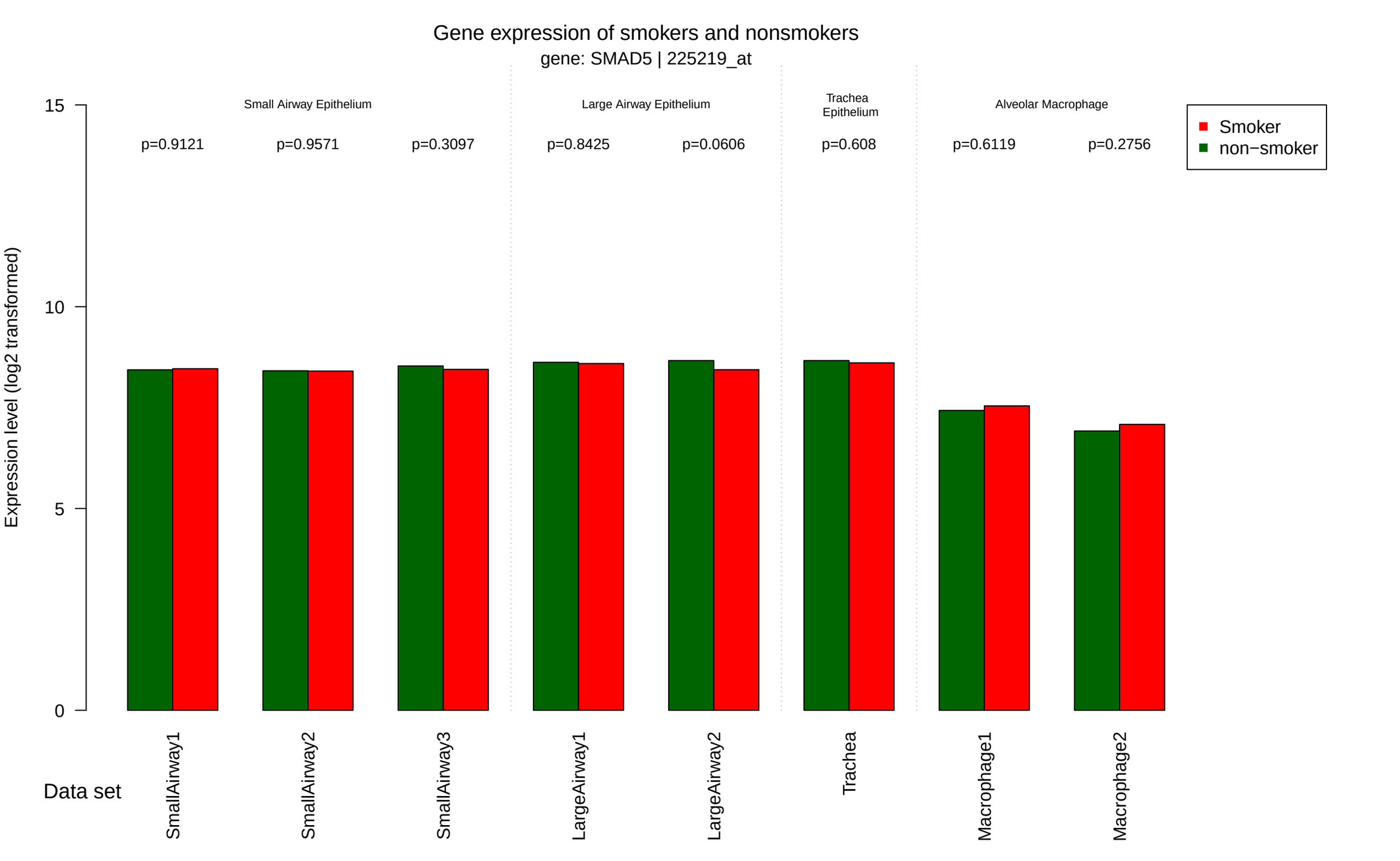
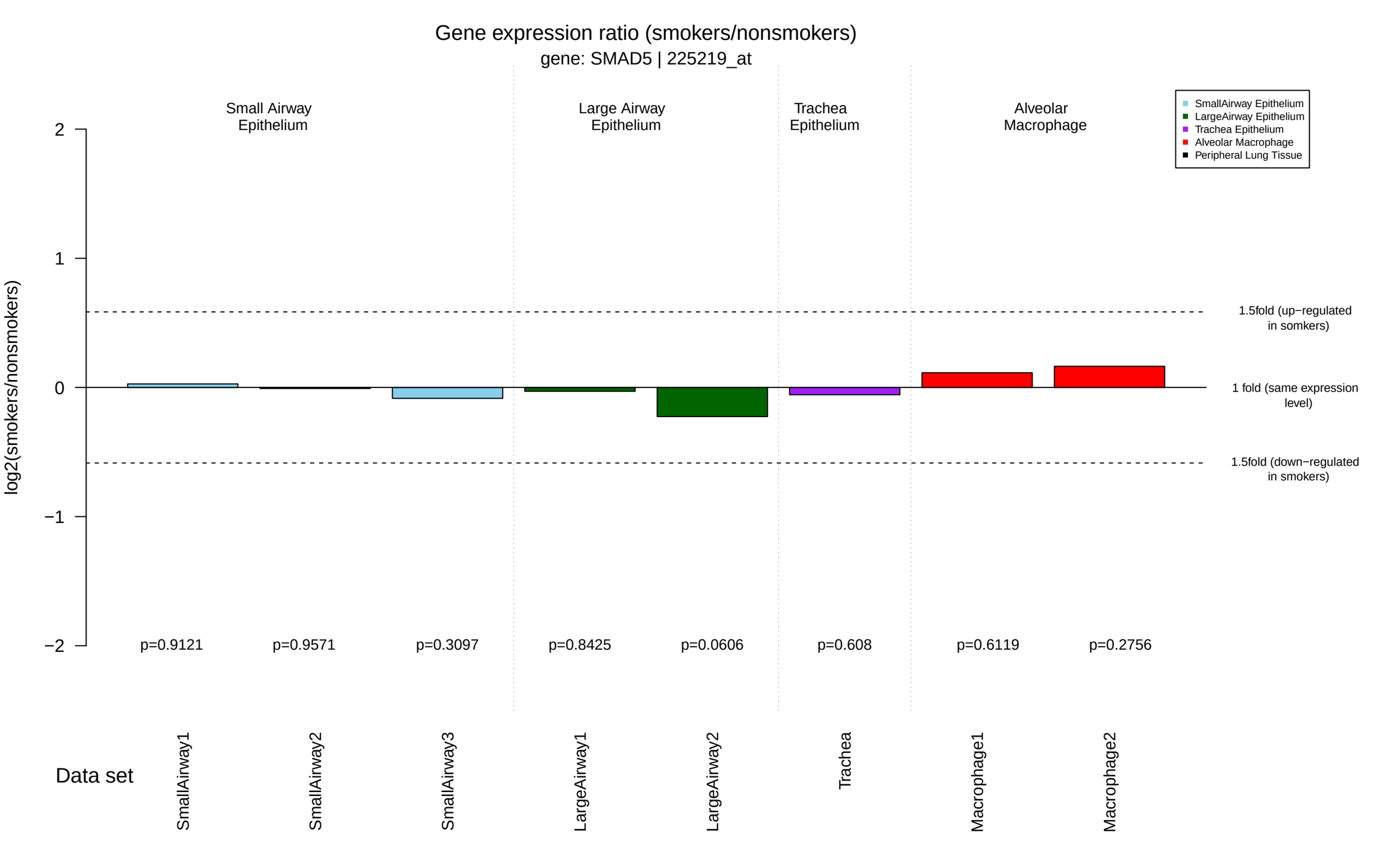
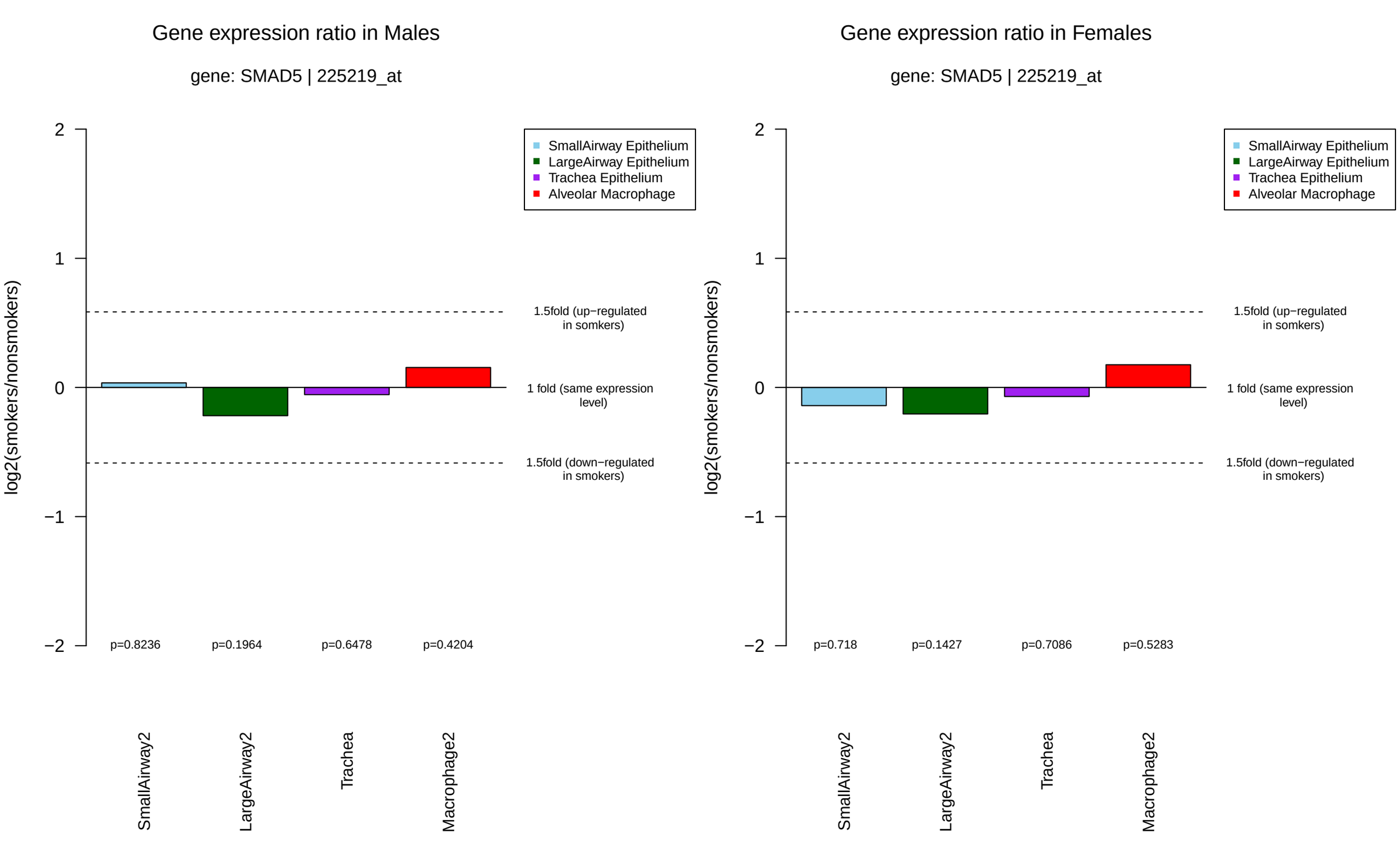
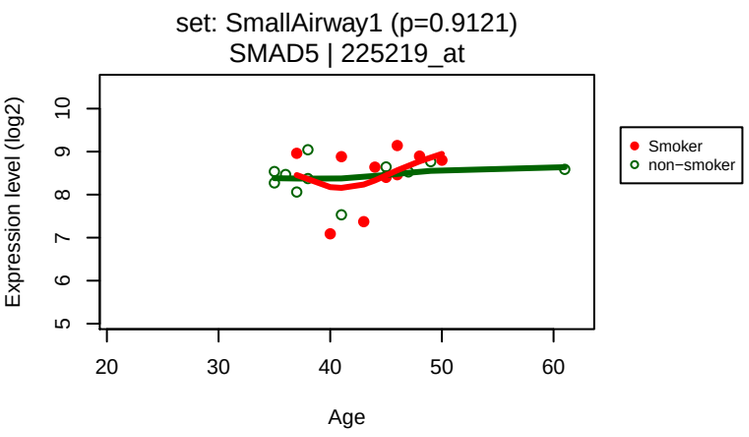
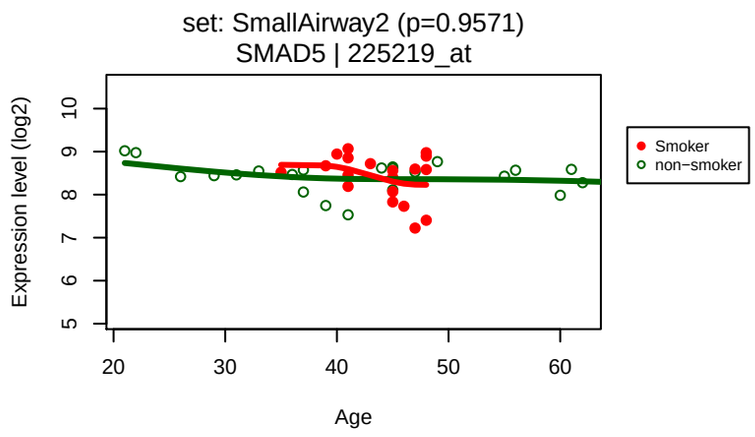
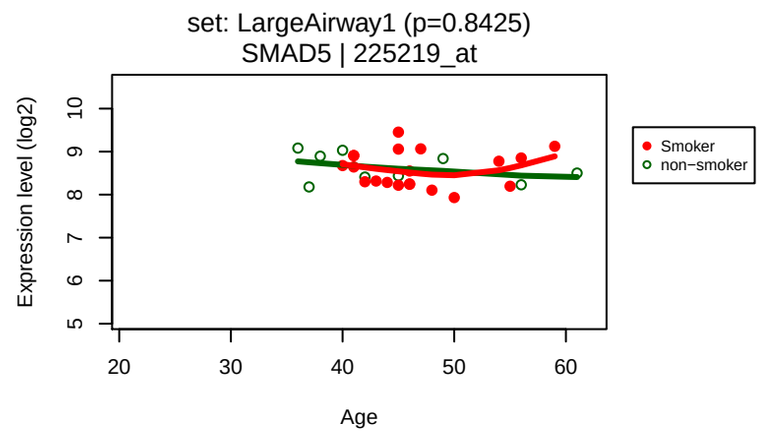

Gene expression of smokers and nonsmokers
gene: SMAD5 | 225219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9121
p=0.9571
p=0.3097
p=0.8425
p=0.0606
p=0.608
p=0.6119
p=0.2756
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

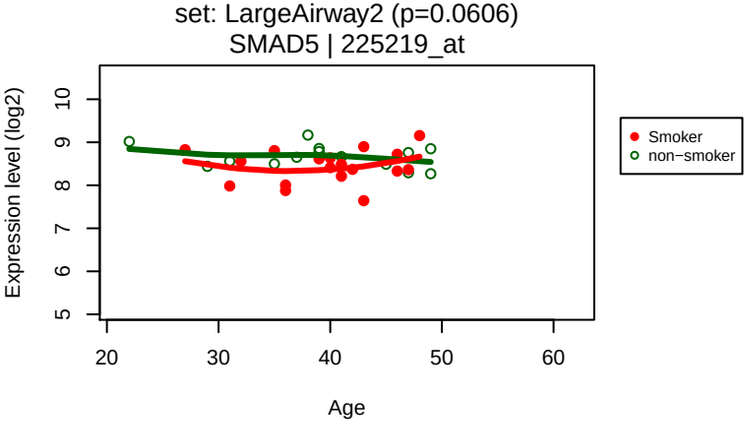

Gene expression ratio (smokers/nonsmokers)
gene: SMAD5 | 225219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9121
p=0.9571
p=0.3097
p=0.8425
p=0.0606
p=0.608
p=0.6119
p=0.2756
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

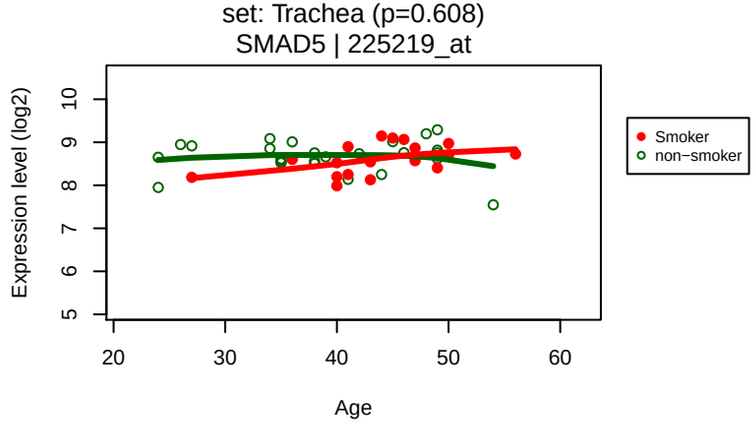

Gene expression ratio in Males
gene: SMAD5 | 225219_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8236
p=0.1964
p=0.6478
p=0.4204
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD5 | 225219_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.718
p=0.1427
p=0.7086
p=0.5283
SmallAirway2
LargeAirway2
Trachea
Macrophage2

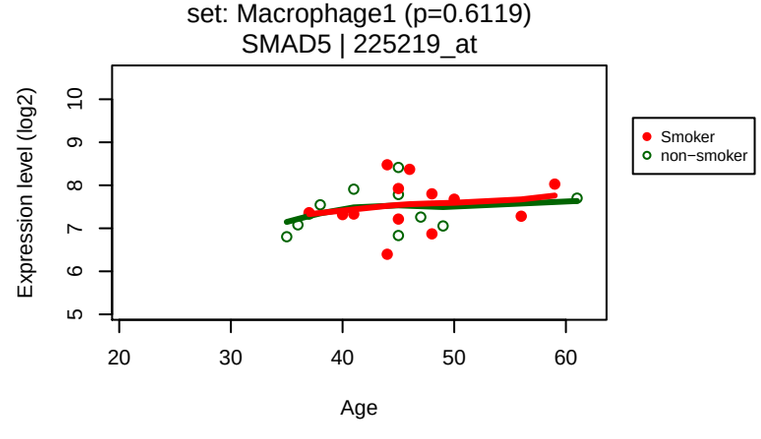

set: SmallAirway1 (p=0.9121)
SMAD5 | 225219_at
Expression level (log2)
Age
Smoker
non−smoker

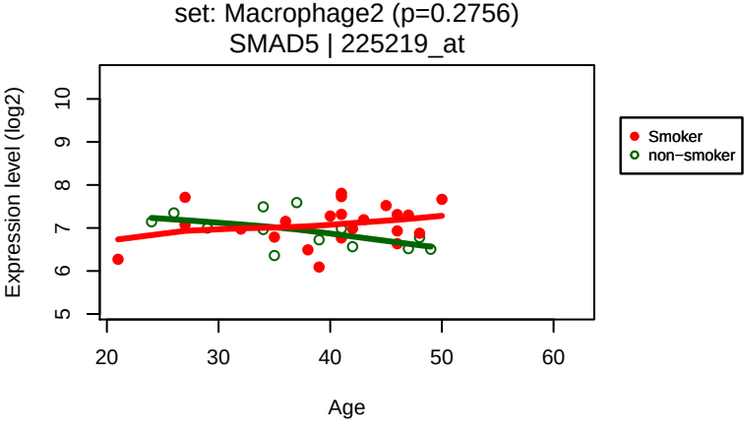

set: SmallAirway2 (p=0.9571)
SMAD5 | 225219_at
Expression level (log2)
Age
Smoker
non−smoker

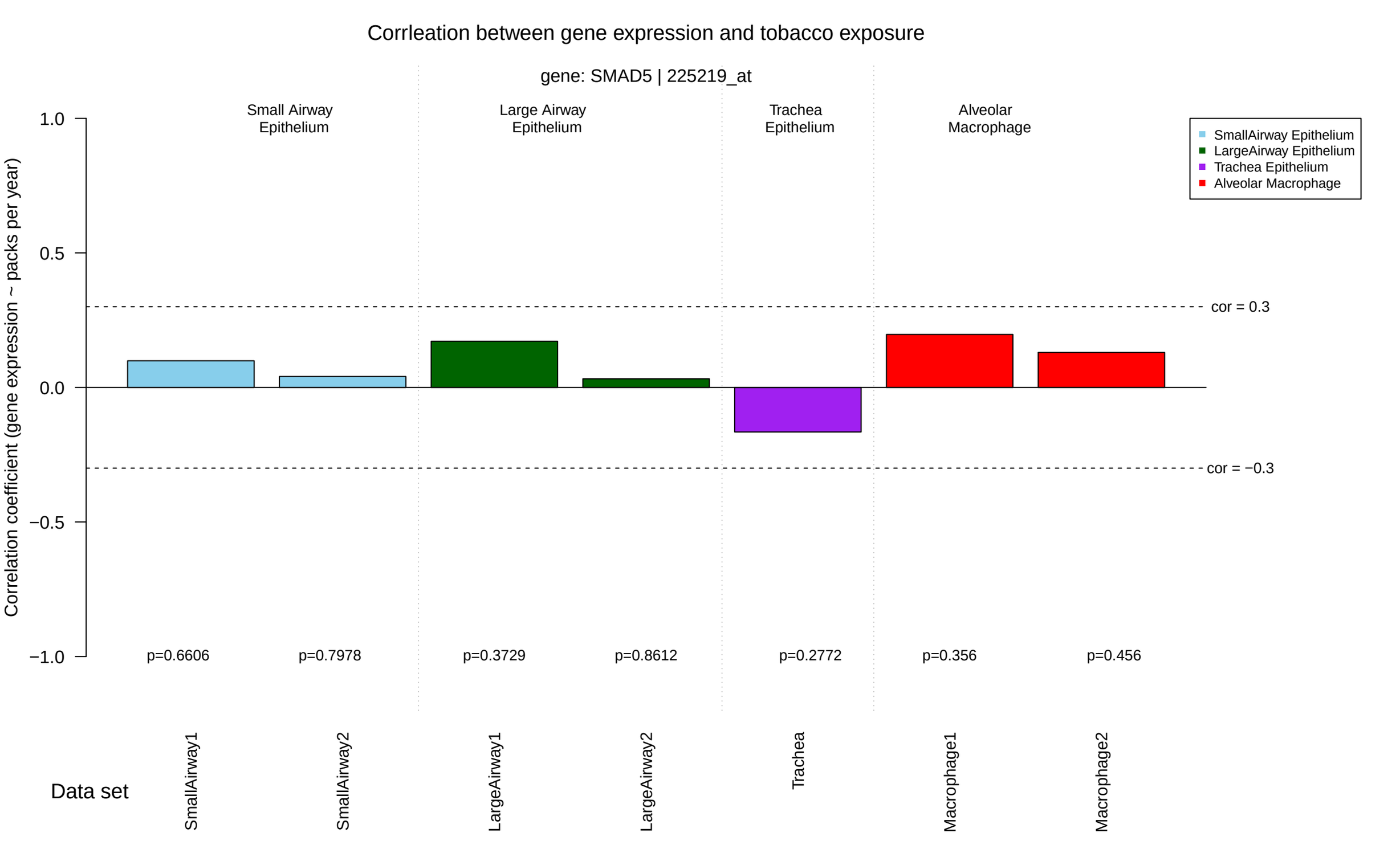

set: LargeAirway1 (p=0.8425)
SMAD5 | 225219_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0606)
SMAD5 | 225219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.608)
SMAD5 | 225219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6119)
SMAD5 | 225219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2756)
SMAD5 | 225219_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD5 | 225219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6606
p=0.7978
p=0.3729
p=0.8612
p=0.2772
p=0.356
p=0.456
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2